

Submission DSA1-MID-SUB01

Question DSA1-MID-Q01

scratch: show step

1. $7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.
 2. Comparison path: $4 \rightarrow 9 \rightarrow 6$; parent of 7 is 6.
- => FINAL: path $4 \rightarrow 9 \rightarrow 6$; parent=6 (check units/interpretation)

Question DSA1-MID-Q02

scratch: show step

1. Maintain a hash set of values seen so far.
 2. Before inserting x , if $x \in \text{seen}$ return True; otherwise insert x .
 3. Expected $O(n)$ time and $O(n)$ auxiliary space; invariant: seen contains exactly the prior prefix.
- => FINAL: hash set membership scan; expected $O(n)$ time, $O(n)$ space; prefix invariant (check units/interpretation)

Question DSA1-MID-Q03

scratch: show step

1. A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.
 2. Height is $\Theta(\log n)$, so search is $\Theta(\log n)$.
- => FINAL: build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree (check units/interpretation)

Question DSA1-MID-Q04

scratch: show step

1. Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.
 2. Order A,B,C,D,E; layer order equals shortest edge distance.
- => FINAL: A,B,C,D,E; BFS explores by distance layers and yields fewest edges (check units/interpretation)

Question DSA1-MID-Q05

scratch: show step

1. Invariant: the prefix before index i is sorted and contains the original prefix values.
 2. Initialization: one-element prefix is sorted; maintenance inserts the next item; termination covers the full array.
- => FINAL: sorted prefix invariant; initialization, insertion maintenance, termination imply correctness (check units/interpretation)

Question DSA1-MID-Q06

scratch: show step

1. $7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.
 2. Comparison path: $4 \rightarrow 9 \rightarrow 6$; parent of 7 is 6.
- => FINAL: path $4 \rightarrow 9 \rightarrow 6$; parent=6 (check units/interpretation)

Question DSA1-MID-Q07

scratch: show step

1. Maintain a hash set of values seen so far.

2. Before inserting x , if $x \in \text{seen}$ return True; otherwise insert x .
 3. Expected $O(n)$ time and $O(n)$ auxiliary space; invariant: seen contains exactly the prior prefix.
- => FINAL: hash set membership scan; expected $O(n)$ time, $O(n)$ space; prefix invariant (check units/interpretation)

Question DSA1-MID-Q08

scratch: show step

1. A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.
 2. Height is $\Theta(\log n)$, so search is $\Theta(\log n)$.
- => FINAL: build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree (check units/interpretation)